Cybersecurity - Undergraduate Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following is not an example of presentation layer issues?
   (a) Poor handling of unexpected input can lead to the execution of arbitrary instructions
   (b) Unintentional or ill-directed use of superficially supplied input
   (c) Cryptographic flaws in the system may get exploited to evade privacy
   (d) Weak or non-existent authentication mechanisms

2. ______________________ is the central node of 802.11 wireless operations.
   (a) WPA
   (b) Access Point
   (c) WAP
   (d) Access Port

3. Which of the following is not a block cipher operating mode?
   (a) ECB
   (b) CFB
   (c) CBF
   (d) CBC

4. When a wireless user authenticates to any AP, both of them go in the course of four-step authentication progression which is called ___________________
   (a) AP-handshaking
   (b) 4-way handshake
   (c) 4-way connection
   (d) wireless handshaking

5. A ________________ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
   (a) front-door
   (b) backdoor
   (c) clickjacking
   (d) key-logging

True / False (10 marks)

Answer True or False

6. True or False: A proxy firewall filters data at the physical layer to control access to network resources.

7. True or False: Public key encryption is faster than symmetric key cryptography, making it the preferred choice for all encryption needs.

8. True or False: A Base Transceiver Station is similar to an Access Point and is used by mobile operators for signal coverage.

9. True or False: A packet filter firewall operates at the application or transport layer, inspecting data for specific applications.

10. True or False: Escaping queries is an effective method to prevent SQL injection attacks by ensuring special characters are treated safely.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find out the number of ways of painting the fence such that at most 2 adjacent posts have the same color for the given fence with n posts and k colors.

12. Write a function to remove the nested record from the given tuple.

End of Paper